ECN 713 Macroeconomic Analysis II
Prof. Natalia Kovrijnykh
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6 Recursive Contracts

This section is based on Chapters 19, 20, and 21 from Ljungqvist and Sargent (2012).

Insurance vs. Incentives

- Risk-averse households facing a stochastic income (endowment) stream
- In the presence of complete markets get full insurance
- We will consider environments where there are incentive problems that make full insurance not achievable (in general)
- Two scenarios
  1. Households cannot commit to contracts
  2. Households actions or incomes are unobservable

We will look at the planner's problem, who designs an efficient contract in the presence of these incentive constraints. The planner will face a trade-off between providing insurance and incentives.

Methodological Theme

- How incentive problems can be managed with *recursive* contracts?
- Contracts issue rewards that depend on the past history. Histories are large-dimensional objects. How to represent histories recursively?

Limited Commitment
This section is based on Thomas and Worrall (1988) and Kocherlakota (1996).

Environment

- There is a large number of ex-ante identical households
- Preferences are given by $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$, u is strictly increasing, strictly concave, C^2
- The endowment stream $\{y_t\}_{t=0}^{\infty}$ is stochastic, where $\forall t \geq 0$ y_t is i.i.d. across time and across households: $Pr\{y_t = \bar{y}_s\} = \pi_s$, $s \in \{1, 2, ..., S\} \equiv \mathcal{S}$, $\bar{y}_{s+1} > \bar{y}_s$.
- History at time t: $y^t = (y_0, y_1, ..., y_t)$
- The lender is risk-neutral
- He has access to perfect credit markets and can borrow or lend at the constant risk-free gross interest rate $R = \beta^{-1}$ (partial equilibrium in this sense).

First Best: no incentive problems

In this case full insurance is achieved. Each HH consumes the per-capita endowment in each period $\sum_{s=1}^{S} \pi_s \bar{y}_s$ and gets a lifetime time utility of $\frac{1}{1-\beta} u(\sum_{s=1}^{S} \pi_s \bar{y}_s)$.

Incentive constraints (to be specified) make this allocation unattainable.

<u>*One-sided Commitment*</u>

The lender can *commit* to contracts, but the HHs cannot. In any period, a HH can walk away from a contract (arrangement with the lender) and choose autarky. Contracts cannot be enforced by the lender or a third party. Contracts must be therefore "self-enforcing".

Autarky

The HH consumes its endowment forever:

$$v_{aut} = E_0 \sum_{t=0}^{\infty} \beta^t u(y_t) = \frac{1}{1-\beta} \sum_{s=1}^{S} \pi_s u(\bar{y}_s)$$

In this case there is no insurance. This is the lowest possible value that the HH can obtain in this economy.

Contracts

A *contract* is a sequence of functions $c_t = f_t(y^t), t \geq 0$, where c_t is the consumption of the household in period t as a function of y^t, the history of income stream up to t. The HH gives y_t to the lender, and receives c_t in exchange.

The payoff to the lender is

$$E_0 \sum_{t=0}^{\infty} \beta^t (y_t - f_t(y^t)),$$

while the payoff to the HH is

$$E_0 \sum_{t=0}^{\infty} \beta^t u(f_t(y^t)).$$

A contract is called *self-enforcing* if $\forall t, \forall y^t$

$$u[f_t(y^t)] + E_t \sum_{j=1}^{\infty} \beta^j u[f_{t+j}(y^{t+j})] \geq u(y_t) + \beta v_{aut}.$$

A contract is called *optimal* (or efficient) if there exists no other self-enforcing contract that offers both parties at least as much expected utility and one party strictly more.

$$P(v_0) = \max_{\{f_t(y^t)\}} E_0 \sum_{t=0}^{\infty} \beta^t (y_t - f_t(y^t))$$

$$\text{s.t.} \qquad E_0 \sum_{t=0}^{\infty} \beta^t u(f_t(y^t)) \geq v_0$$

$$u[f_t(y^t)] + E_t \sum_{j=1}^{\infty} \beta^j u[f_{t+j}(y^{t+j})] \geq u(y_t) + \beta v_{aut} \ \forall t, \forall y^t$$

$P(v_0)$ defines a constrained Pareto frontier as a function of v_0.

How to represent this recursively?
Trick: an efficient contract cannot be Pareto dominated (along the equilibrium path) after any history. If it were, it would be possible to replace that part of the contract by simultaneously relaxing all previous self-enforcing constraints. The new contract would then be self-enforcing and dominate the old one at date $t = 0$.

The Pareto frontier after y^t is also given by P. What matters is where on the frontier then lender and the HH are at any given period $(v_t, P(v_t))$. Essentially, the value to the HH at time t ("promised value") summarizes the history up to date t.

Recursive formulation

Following Spear and Srivastava (1987), letting v be the promised value to the HH, we can write the problem recursively as follows:

$$
\begin{aligned}
& P(v) = \max_{\{c_s, w_s\}} \sum_{s=1}^{S} \pi_s[\bar{y}_s - c_s + \beta P(w_s)] \\
\text{s.t.} \quad & \sum_{s=1}^{S} \pi_s[u(c_s) + \beta w_s] \geq v, \qquad (PK)\ (\mu) \\
& u(c_s) + \beta w_s \geq u(\bar{y}_s) + \beta v_{aut} \quad \forall s, \quad (PC)\ (\lambda_s) \\
& c_s \in [c_{min}, c_{max}], \\
& w_s \in [v_{aut}, \bar{v}].
\end{aligned}
$$

- $P(v)$ is decreasing in v
- $P(v)$ is concave in v because the constraint set is convex and the one-period payoff of the lender is (weakly) concave
- We can show that in fact $P(v)$ is strictly concave and also that $P(v)$ is differentiable (see LS)

The full-commitment benchmark of this problem is as follows

$$
P^*(v) = \max_{\{c_s, w_s\}} \sum_{s=1}^{S} \pi_s[\bar{y}_s - c_s + \beta P^*(w_s)]
$$

$$
\text{s.t.} \qquad \sum_{s=1}^{S} \pi_s[u(c_s) + \beta w_s] \geq v. \qquad (PK)\ (\mu)
$$

The Envelope and F.O.C. of this problem are

$$
P^{*\prime}(w_s) = -\mu,
$$

$$
\mu u'(c_s) = 1,
$$

$$
P^{*\prime}(v) = -\mu
$$

implying

$$u'(c_s) = \frac{1}{\mu},$$

$$P^{*\prime}(v) = P^{*\prime}(w_s).$$

Hence, we can write

$$P^*(v) = \underbrace{\sum_{s=1}^{S} \pi_s \bar{y}_s}_{Ey} - c(v) + \beta P^*(v),$$

$$\sum_{s=1}^{S} \pi_s u(c(v)) + \beta v = v \Rightarrow u(c(v)) = (1-\beta)v \Rightarrow c(v) = u^{-1}((1-\beta)v).$$

Substituting the last expression in the previous one

$$(1-\beta)P^*(v) = Ey - u^{-1}((1-\beta)v) \Rightarrow P^*(v) = \frac{1}{1-\beta}[Ey - \underbrace{u^{-1}((1-\beta)v)}_{c(v)}].$$

If $P^*(v) = 0$, then $\underbrace{u^{-1}((1-\beta)v}_{c(v)} = Ey$.

Back to the limited-commitment problem

Lagrangian

$$\mathcal{L} = \sum_{s=1}^{S} \pi_s[\bar{y}_s - c_s + \beta P(w_s)] + \mu \Big\{ \sum_{s=1}^{S} \pi_s [u(c_s) + \beta w_s] - v \Big\}$$
$$+ \sum_{s=1}^{S} \lambda_s \Big\{ u(c_s) + \beta w_s - [u(\bar{y}_s) + \beta v_{aut}] \Big\}$$

F.O.C.

$$[c_s]: \ (\lambda_s + \mu \pi_s) u'(c_s) = \pi_s \tag{1}$$
$$[w_s]: \ \lambda_s + \mu \pi_s = -\pi_s P'(w_s) \tag{2}$$

Envelope

$$P'(v) = -\mu \tag{3}$$

Combining (1) and (2) we have

$$u'(c_s) = -\frac{1}{P'(w_s)}, \tag{4}$$

which gives an upward-sloping relationship between c_s and w_s, since u and P are both concave. Notice that the optimal contract equates the HH's MRS between c_s and w_s, $u'(c_s)/\beta$, with the lender's MRT, $-1/\beta P'(w_s)$.

Combining (2) and (3) we get

$$\begin{aligned} P'(w_s) &= -\mu - \frac{\lambda_s}{\pi_s} \\ &= P'(v) - \frac{\lambda_s}{\pi_s} \end{aligned} \tag{5}$$

(Note: The first-best corresponds to $\lambda_s = 0 \ \forall s \Rightarrow c_s = c \ \forall s, \ w_s = v \ \forall s \Rightarrow$ full insurance.)

How w_s varies with v depends on wether the income realization $\bar{y}_s$ is such that PC binds ($\lambda_s > 0$) or not ($\lambda_s = 0$).

<u>States where $\lambda_s > 0$</u>

- $\lambda_s > 0 \Rightarrow$ PC holds with equality: $u(c_s) + \beta w_s = u(\bar{y}_s) + \beta v_{aut}$.
- $\lambda_s > 0$ in (5) $\Rightarrow P'(w_s) < P'(v) \Rightarrow w_s > v$ (because P is concave)

$\Rightarrow c_s < \bar{y}_s$, since $w_s > v \geq v_{aut}$.

I.e., when PC binds, the lender induces the HH to consume less than its endowment today by raising its continuation value.

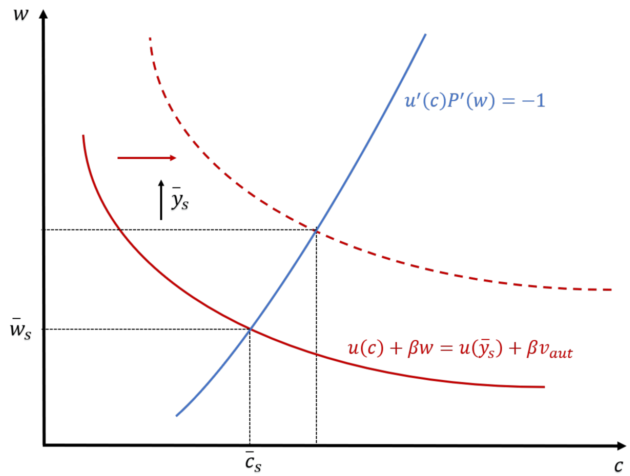
Intuitively, insurance against income risk means in some states $c_s \geq \bar{y}_s$ (borrow) and in other states $c_s < \bar{y}_s$ (repay). Therefore, HHs will borrow when $\bar{y}_s$ is low and repay when it is high.

When $\bar{y}_s$ is high, there is temptation to walk away. To provide incentives to the HH stay in the relationship, the lender increases its continuation value.

When $\lambda_s > 0$, we have

$$\left.\begin{aligned} &u(c_s) + \beta w_s = u(\bar{y}_s) + \beta v_{aut} \\ &u'(c_s) = -P'(w_s)^{-1} \end{aligned}\right\} \text{two equations in } w_s, c_s, \text{ independent of } v$$

Figure 1: One-sided Commitment: Optimal Contract



Kocherlakota (1996) calls this "amnesia": when income realization is such that PC binds, the optimal contract disposes of all history dependence, and makes both consumption and continuation values depend only on the current income.

$$c(v, y) = g_1(y)$$

$$w(v, y) = h_1(y)$$

Then, the contract in this case can be summarized as

$$(\bar{c}_s, \bar{w}_s) = (g_1(\bar{y}_s), h_1(\bar{y}_s)).$$

States where $\lambda_s = 0$

When $\lambda_s = 0$ in (5) $\Rightarrow P'(v) = P'(w_s) \Rightarrow w_s = v$. Also, substituting in (4) we have that $u'(c_s) = -1/P'(v)$, consumption in state s depends on v but not on $\bar{y}_s$.

$$c(v, y) = g_2(v)$$

$$w(v, y) = v$$

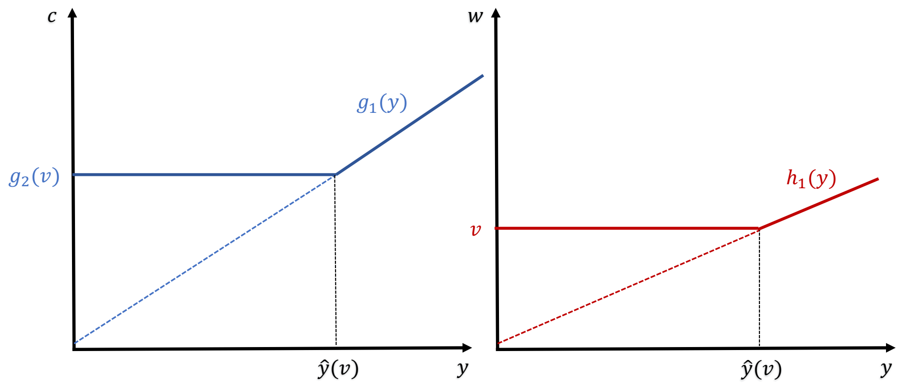
The optimal contract

Combining the two cases,

$$c(v, y) = \max\{g_1(y), g_2(v)\}$$

$$w(v, y) = \max\{h_1(y), v\}$$

Figure 2: One-sided Commitment: Optimal Contract



This implies that there is a cut-off value $\hat{y}(v)$ such that $\lambda_s > 0 \iff \bar{y}_s > \hat{y}(v)$. This can be found from

$$u'(g_2(v)) = -\frac{1}{P'(v)}$$

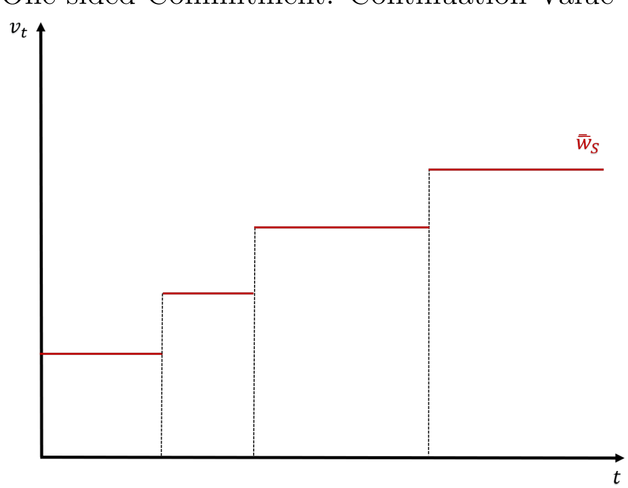
$$u[\hat{y}(v)] = u[g_2(v)] + \beta(v - v_{aut})$$

and by concavity of u and P, $\hat{y}(v)$ is increasing in v.

We have that $v_{t+1} = w(v_t, y_t)$, $c_t = c(v_t, y_t)$. As $\{y_t\}_{t=0}^{\infty}$ is a stochastic process so are $\{v_{t+1}\}_{t=0}^{\infty}, \{c_t\}_{t=0}^{\infty}$.

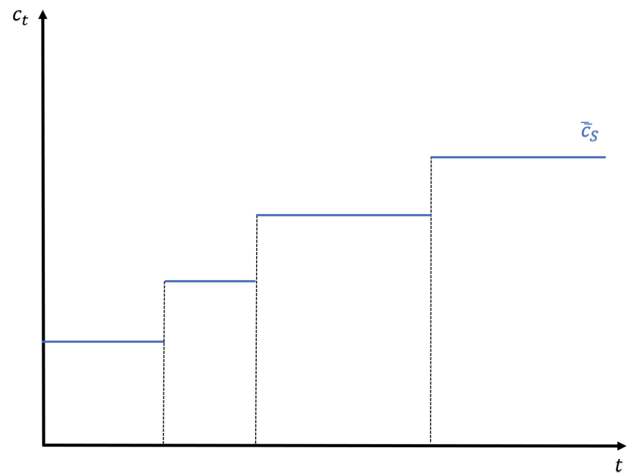
If $y_t \leq \hat{y}(v_t)$ then $v_{t+1} = v_t$ and if $y_t > \hat{y}(v_t)$ then $v_{t+1} > v_t$. Hence, the promised values never decrease.

Figure 3: One-sided Commitment: Continuation Value Over Time



Suppose $\bar{y}_S$ is realized at any promised value $v \Rightarrow$ jump to $\bar{w}_S, \bar{c}_S$, independent of v. For any income realization thereafter, the continuation value and consumption are equal to $\bar{w}_S, \bar{c}_S$. For $(\bar{w}_S, \bar{c}_S)$ we have $\bar{w}_S = \frac{u(\bar{c}_S)}{1-\beta}$ and $u(\bar{c}_S) + \beta\bar{w}_S = u(\bar{y}_S) + \beta v_{aut}$.

Figure 4: One-sided Commitment: Consumption Over Time



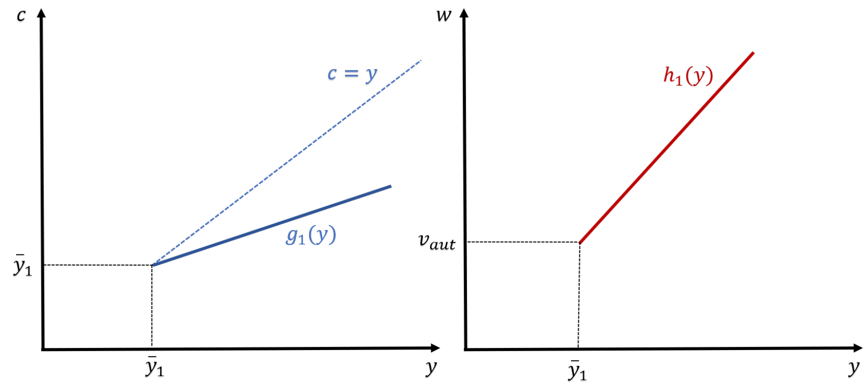
Suppose we start with a promised value of v_{aut}. Working backwards for $j = S-1, ..., 1$ compute $\bar{c}_j, \bar{w}_j$:

$$u(\bar{c}_j) + \beta\bar{w}_j = u(\bar{y}_j) + \beta v_{aut}$$

$$\bar{w}_j = [u(\bar{c}_j) + \beta\bar{w}_j] \sum_{k=1}^{j} \pi_k + \sum_{k=j}^{S} \pi_k [u(\bar{c}_k) + \beta\bar{w}_k]$$

Can verify (see LS) that $\bar{c}_1 = \bar{y}_1$, $\bar{w}_1 = v_{aut} \Rightarrow \hat{y}(v_{aut}) = \bar{y}_1$. Moreover, $\bar{c}_j < \bar{y}_j$ for $j \geq 2$. At the first period the HH never borrows. As time passes it will borrow at low income realizations and repays at high income realizations.

Figure 5: One-sided Commitment: First Period Consumption and Continuation Values



Summarize

Promised values and consumption never decrease. Eventually (after $\bar{y}_S$ is realized for the first time) full insurance is attained: HH receive $\bar{c}_S$ in each period and the continuation value is $\bar{w}_S = u(\bar{c}_S)/(1-\beta)$. Whether $\bar{c}_S \lesseqgtr \sum_{j=1}^{S} \pi_j \bar{y}_j$ depends on whether $P(\bar{w}_S) = \sum_{j=1}^{S} \pi_j \frac{\bar{y}_j - \bar{c}_S}{1-\beta} \lesseqgtr 0$. We can show that $P(v_{aut}) > 0$ and that the value for the lender looks as below.

What if the lender can walk away? Then we would need to impose the following additional constraints:

$$\bar{y}_s - c_s + \beta P(w_s) \geq 0 \ \forall s,$$
$$P(w_s) \geq 0 \ \forall s.$$

Exercise. *Suppose that neither the HH nor the lender can commit to contracts. Show that* $\exists \beta^* \in (0,1)$ *such that*

- *for* $\beta < \beta^*$ *full insurance is not a self-enforcing SPE*
- $\beta \geq \beta^*$ *full insurance is a self-enforcing SPE*

This means that for high enough β we can sustain full insurance even if the lender can walk away.

Many Households

At $t = 0$ offer the lender offers each HH v_{aut}. At first, the cross-sectional distribution of consumer and continuation values will "fan out" and then it will "fan in". Eventually, this distribution asymptotically becomes concentrated at $\bar{c}_S = g_2(\bar{v})$, where $\bar{v} = \bar{w}_S$ and $\lim_{t\to\infty} \Pr(c_t = \bar{c}_S) = 1$.

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